



Label claim
Each film coated tablet contains:
Efavirenz 600 mg
Efavirenz Tablets 600 mg: Peach coloured capsule shaped, film coated tablets, debossed with 'M109' on one side and plain on the other side.
Dosage Form: Oral, Film coated tablet
Therapeutic indications
Efavirenz is indicated in antiviral combination treatment of HIV-1 infected adults, adolescents and children.

Posology and method of administration
The therapy should be initiated by a physician experienced in the management of HIV infection.
Concomitant antiretroviral therapy: Efavirenz must be given in combination with other antiretroviral medications.
Efavirenz may be taken with or without food. It is recommended that Efavirenz tablet be taken on an empty stomach. The increased efavirenz concentrations observed following administration of Efavirenz with food may lead to an increase in frequency of adverse events.
In order to improve the tolerability of nervous system undesirable effects, bedtime dosing is recommended during the first 2-4 weeks of therapy and in patients who continue to experience adverse events.
Adults: The recommended dosage of Efavirenz in combination with nucleoside analogue reverse transcriptase inhibitors (NRTIs) with or without a protease inhibitor (PI) is 600 mg orally, once daily.
Efavirenz film-coated tablets are not suitable for children <3years or weighing less than 40 kg.
Contraindications
Hypersensitivity to Efavirenz or to any of the excipients.
Efavirenz must not be used in patients with severe hepatic impairment (Child Pugh Grade C).
Efavirenz must not be administered concurrently with terfenadine, astemizole, cisapride, midazolam, triazolam, pimozone, bepridil, or ergot alkaloids (for example, ergotamine, dihydroergotamine, ergonovine, and methylergonovine) because competition for CYP3A4 by efavirenz could result in inhibition of metabolism and create the potential for serious and/or life-threatening undesirable effects (for example, cardiac arrhythmias, prolonged sedation or respiratory depression).
Herbal preparations containing St. John's wort (*Hypericum perforatum*) must not be used while taking efavirenz due to the risk of decreased plasma concentrations and reduced clinical effects of efavirenz.

Special warning and precautions for use
Efavirenz must not be used as a single agent to treat HIV or added on as a sole agent to a failing regimen. As with all other non-nucleoside reverse transcriptase inhibitors (NRTIs), resistant virus emerges rapidly when efavirenz is administered as monotherapy. The choice of new antiretroviral agent(s) to be used in combination with efavirenz should take into consideration the potential for viral cross-resistance. Co-administration of efavirenz with the fixed combination tablet containing efavirenz, emtricitabine, and tenofovir disoproxil fumarate, is not recommended.
When prescribing medicinal products concomitantly with Efavirenz, physicians should refer to the corresponding Summary of Product Characteristics.
QTC prolongation has been observed with the use of efavirenz (see Section Pharmacodynamics and Section Interaction with Other Medicaments). Consider alternatives to Efavirenz Tablets when coadministered with a drug with a known risk of Torsade de Pointes or when administered to patients at higher risk of Torsade de Pointes.
Patients should be advised that current antiretroviral therapy, including efavirenz, has not been proven to prevent the risk of transmission of HIV to others through sexual contact or blood contamination. Appropriate precautions should continue to be employed.
If any antiretroviral medicinal product in a combination regimen is interrupted because of suspected intolerance, serious consideration should be given to simultaneous discontinuation of all antiretroviral medicinal products. The antiretroviral medicinal products should be restarted at the same time upon resolution of the intolerance symptoms. Intermittent monotherapy and sequential reintroduction of antiretroviral agents is not advisable because of the increased potential for selection of resistant virus.
Rash: Mild to moderate rash has been reported in clinical studies with efavirenz and usually resolves with continued therapy. Appropriate antihistamines and/or corticosteroids may improve the tolerability and hasten the resolution of rash. Severe rash associated with blistering, moist desquamation or ulceration has been reported in less than 1% of patients treated with efavirenz. The incidence of erythema multiforme or Stevens Johnson syndrome was approximately 0.1%. Efavirenz must be discontinued in patients developing severe rash associated with blistering, desquamation, mucosal involvement or fever. If therapy with efavirenz is discontinued, consideration should also be given to interrupting therapy with other antiretroviral agents to avoid development of resistant virus.
Prophylaxis with appropriate antihistamines prior to initiating therapy with efavirenz in children may be considered.
Patients who discontinued treatment with other NRTIs due to rash may be at higher risk of developing rash during treatment with efavirenz.

Psychiatric symptoms: Psychiatric adverse experiences have been reported in patients treated with efavirenz. Patients with a prior history of psychiatric disorders appear to be at greater risk of these serious psychiatric adverse experiences. In particular, severe depression was more common in those with a history of depression. There have also been post-marketing reports of severe depression, death by suicide, delusions and psychosis like behaviour. Patients should be advised that if they experience symptoms such as severe depression, psychosis or suicidal ideation, they should contact their doctor immediately to assess the possibility that the symptoms may be related to the use of efavirenz, and if so, to determine whether the risks of continued therapy outweigh the benefits.

Nervous system symptoms: Symptoms including, but not limited to, dizziness, insomnia, somnolence, impaired concentration and abnormal dreaming are frequently reported undesirable effects in patients receiving efavirenz 600 mg daily in clinical studies. Nervous system symptoms usually begin during the first one or two days of therapy and generally resolve after the first 2 to 4 weeks. Patients should be informed that if they do occur, these common symptoms are likely to improve with continued therapy and are not predictive of subsequent onset of any of the less frequent psychiatric symptoms.

Late-onset neurotoxicity, including ataxia and encephalopathy (impaired consciousness, confusion, psychomotor slowing, psychosis, delirium), may occur months to years after beginning efavirenz therapy. Some events of late-onset neurotoxicity have occurred in patients with CYP2B6 genetic polymorphisms, which are associated with increased efavirenz levels despite standard dosing of Efavirenz Tablets. Patients presenting with signs and symptoms of serious neurologic adverse experiences should be evaluated promptly to assess the possibility that these events may be related to efavirenz use, and whether discontinuation of Efavirenz Tablets is warranted.

Seizures: Convulsions have been observed rarely in patients receiving efavirenz, generally in the presence of known medical history of seizures. Patients who are receiving concomitant anticonvulsant medicinal products primarily metabolised by the liver, such as phenytoin, carbamazepine and phenobarbital, may require periodic monitoring of plasma levels. In a drug interaction study, carbamazepine plasma concentrations were decreased when carbamazepine was co-administered with efavirenz. Caution must be taken in any patient with a history of seizures.

Effect of food: The administration of Efavirenz with food may increase efavirenz exposure and may lead to an increase in the frequency of undesirable effects. This effect may be more evident for the film-coated tablets than for the hard capsules. It is recommended that Efavirenz be taken on an empty stomach, preferably at bedtime.

Immune Reactivation Syndrome: In HIV infected patients with severe immune deficiency at the time of institution of combination antiretroviral therapy (cART), an inflammatory reaction to asymptomatic or residual opportunistic pathogens may arise and cause serious clinical conditions, or aggravation of symptoms. Typically, such reactions have been observed within the first few weeks or months of initiation of cART. Relevant examples are cytomegalovirus retinitis, generalised and/or local mycobacterial infections, and pneumonia caused by *Pneumocystis jirovecii* (formerly known as *Pneumocystis carinii*). Any inflammatory symptoms should be evaluated and treatment instituted when necessary.

Lipodystrophy and metabolic abnormalities: Combination antiretroviral therapy has been associated with the redistribution of body fat (lipodystrophy) in HIV patients. The long-term consequences of these events are currently unknown. Knowledge about the mechanism is incomplete. A connection between visceral lipomatosis and PIs and lipodystrophy and NRTIs has been hypothesised. A higher risk of lipodystrophy has been associated with individual factors such as older age, and with drug related factors such as longer duration of antiretroviral treatment and associated metabolic disturbances. Clinical examination should include evaluation for physical signs of fat redistribution. Consideration should be given to the measurement of fasting serum lipids and blood glucose. Lipid disorders should be managed as clinically appropriate.

Osteonecrosis: Although the etiology is considered to be multifactorial (including corticosteroid use, alcohol consumption, severe immunosuppression, higher body mass index), cases of osteonecrosis have been reported particularly in patients with advanced HIV-disease and/or long-term exposure to combination antiretroviral therapy (cART). Patients should be advised to seek medical advice if they experience joint aches and pain, joint stiffness or difficulty in movement.
Lactose: This medicinal product contains 256 mg of lactose in 600 mg in each daily dose respectively. This quantity is not likely to induce symptoms of lactose intolerance.

Efavirenz film-coated tablets are unsuitable for individuals with the rare hereditary disorders of galactosaemia or glucose/galactose malabsorption syndrome. Individuals with these conditions may take efavirenz oral solution, which is free from lactose.

Special populations:
Liver disease: Because of the extensive cytochrome P450-mediated metabolism of efavirenz and limited clinical experience in patients with chronic liver disease, caution must be exercised in administering efavirenz to patients with mild-to-moderate liver disease. Patients should be monitored carefully for dose-related adverse events, especially nervous system symptoms. Laboratory tests should be performed to evaluate their liver disease at periodic intervals.

The safety and efficacy of efavirenz has not been established in patients with significant underlying liver disorders. Efavirenz is contraindicated in patients with severe hepatic impairment. Patients with chronic hepatitis B or C and treated with combination antiretroviral therapy are at increased risk for severe and potentially fatal hepatic adverse events. Patients with pre-existing liver dysfunction including chronic active hepatitis have an increased frequency of liver function abnormalities during combination antiretroviral therapy and should be monitored according to standard practice. If there is evidence of worsening liver disease or persistent elevations of serum transaminases to greater than 5 times the upper limit of the normal range, the benefit of continued therapy with efavirenz needs to be weighed against the potential risks of significant liver toxicity. In such patients, interruption or discontinuation of treatment must be considered.

In patients treated with other medicinal products associated with liver toxicity, monitoring of liver enzymes is also recommended. In case of concomitant antiviral therapy for hepatitis B or C, please refer also to the relevant product information for these medicinal products.
Renal insufficiency: The pharmacokinetics of efavirenz have not been studied in patients with renal insufficiency; however, less than 1% of an efavirenz dose is excreted unchanged in the urine, so the impact of renal impairment on efavirenz elimination should be minimal. There is no experience in patients with severe renal failure and close safety monitoring is recommended in this population.

Elderly: Insufficient numbers of elderly patients have been evaluated in clinical studies to determine whether they respond differently than younger patients.

Children: Efavirenz has not been evaluated in children below 3 years of age or who weigh less than 13 kg.

Interaction with other medicinal products and other forms of interaction

Efavirenz is an inducer of CYP3A4 and an inhibitor of some CYP isozymes including CYP3A4. Other compounds that are substrates of CYP3A4 may have decreased plasma concentrations when co-administered with efavirenz. Efavirenz exposure may also be altered when given with medicinal products or food (for example, grapefruit juice) which affect CYP3A4 activity.

QT Prolonging Drugs

There is limited information available on the potential for a pharmacodynamic interaction between Efavirenz Tablets and drugs that prolong the QTc interval. QTc prolongation has been observed with the use of efavirenz (see Section Pharmacodynamics and Section Warnings and Precautions). Consider alternatives to Efavirenz Tablets when coadministered with a drug with a known risk of Torsade de Pointes.

Contraindications of concomitant use

Efavirenz must not be administered concurrently with terfenadine, astemizole, cisapride, midazolam, triazolam, pimozone, bepridil, or ergot alkaloids (for example, ergotamine, dihydroergotamine, ergonovine, and methylergonovine) since inhibition of their metabolism may lead to serious, life-threatening events.

St. John's wort (*Hypericum perforatum*): Co-administration of efavirenz and St. John's wort or herbal preparations containing St. John's wort is contraindicated. Plasma levels of efavirenz can be reduced by concomitant use of St. John's wort due to induction of drug metabolising enzymes and/or transport proteins by St. John's wort. If a patient is already taking St. John's wort, stop St. John's wort, check viral levels and if possible efavirenz levels. Efavirenz levels may increase on stopping St. John's wort and the dose of efavirenz may need adjusting. The inducing effect of St. John's wort may persist for at least 2 weeks after cessation of treatment.

Other interactions

Interactions between efavirenz and protease inhibitors, antiretroviral agents other than protease inhibitors and other non-antiretroviral medicinal products are listed in Table 1 below (increase is indicated as "↑", decrease as "↓", no change as "↔", and once every 8 or 12 hours as "q8h" or "q12h"). If available, 90% or 95% confidence intervals are shown in parentheses. Studies were conducted in healthy subjects unless otherwise noted.

Table 1: Interactions between efavirenz and other medicinal products

Medicinal product by therapeutic areas (dose)	Effects on drug levels C_{max} , C_{min} with confidence intervals if available ^a (mechanism)	Recommendation concerning co-administration with efavirenz
ANTI-INFECTIVES		
Antiretrovirals		
Protease inhibitors (PI)		
Atazanavir / ritonavir / Efavirenz (400 mg once daily/100 mg once daily/ 600 mg once daily, all administered with food)	Atazanavir (pm): AUC: ↔* (↓9% to ↑10%) C_{min} : ↑17%* (↑8% to ↑27%) C_{min} : ↓42%* (↓31% to ↓51%) Atazanavir (pm): AUC: ↔*/*** (↓10% to ↑26%) C_{min} : ↔*/*** (↓5% to ↑26%) C_{min} : ↑12%*/* (↓16% to ↑49%) (CYP3A4 induction). *When compared to atazanavir 300 mg/ritonavir 100 mg once daily in the evening without efavirenz. This decrease in atazanavir C_{min} might negatively impact the efficacy of atazanavir. ** based on historical comparison	Co-administration of efavirenz with atazanavir / ritonavir is not recommended. If the co-administration of atazanavir with an NRTI is required, an increase in the dose of both atazanavir and ritonavir to 400 mg and 200 mg, respectively, in combination with efavirenz could be considered with close clinical monitoring.
Fosamprenavir / ritonavir / Efavirenz (700 mg twice daily/100 mg twice daily/ 600 mg once daily)	No clinically significant pharmacokinetic interaction	No dosage adjustment is necessary for any of these medicinal products. See also ritonavir row below.
Fosamprenavir / Nelfinavir / Efavirenz	Interaction not studied.	No dosage adjustment is necessary for any of these medicinal products.
Fosamprenavir / Saquinavir / Efavirenz	Interaction not studied.	Not recommended as the exposure to both PIs is expected to be significantly decreased.
Indinavir / Efavirenz (800 mg q8h / 200 mg once daily)	Indinavir: AUC: ↓ 31% (↓ 8% to ↓ 47%) C_{min} : ↓ 40% A similar reduction in indinavir exposures was observed when indinavir 1000 mg q8h was given with efavirenz 600 mg daily. (CYP3A4 induction) Efavirenz: No clinically significant pharmacokinetic interaction	While the clinical significance of decreased indinavir concentrations has not been established, the magnitude of the observed pharmacokinetic interaction should be taken into consideration when choosing a regimen containing both efavirenz and indinavir.
Indinavir / ritonavir / Efavirenz (800 mg twice daily /100 mg twice daily / 600 mg once daily)	Indinavir: AUC: ↓ 25% (↓ 16% to ↓ 32%) ^b C_{min} : ↓ 17% (↓ 6% to ↓ 26%) ^b C_{min} : ↓ 50% (↓ 40% to ↓ 59%) ^b Efavirenz: No clinically significant pharmacokinetic interaction. The geometric mean C_{min} for indinavir (0.33 mg/l) when given with ritonavir and efavirenz was higher than the mean historical C_{min} (0.15 mg/l) when indinavir was given alone at 800 mg q8h. In HIV-1 infected patients (n = 6), the pharmacokinetics of indinavir and efavirenz were generally comparable to these uninfected volunteer data.	No dosage adjustment is necessary for efavirenz when given with indinavir or indinavir/ritonavir. See also ritonavir row below.
Lopinavir / ritonavir soft capsules or oral solution / Efavirenz Lopinavir / ritonavir tablets / Efavirenz (400 / 100 mg twice daily / 600 mg once daily) (500 / 125 mg twice daily / 600 mg once daily)	Substantial decrease in lopinavir exposure. Lopinavir concentrations: ↓ 30-40% Lopinavir concentrations: Similar to lopinavir/ritonavir 400/100 mg twice daily without efavirenz	With efavirenz, an increase of the lopinavir/ritonavir soft capsule or oral solution doses by 33% should be considered (4 capsules/ – 6.5 ml twice daily instead of 3 capsules/ 5 ml twice daily). Caution is warranted since this dosage adjustment might be insufficient in some patients. The dosage of lopinavir / ritonavir tablets should be increased to 500/125 mg twice daily when co-administered with efavirenz 600 mg once daily. See also ritonavir row below.
Nelfinavir / Efavirenz (750 mg q8h / 600 mg once daily)	Nelfinavir: AUC: ↑ 20% (↑ 8% to ↑ 34%) C_{min} : ↑ 21% (↑ 10% to ↑ 33%) The combination was generally well tolerated.	No dosage adjustment is necessary for either medicinal product.
Ritonavir / Efavirenz (500 mg twice daily / 600 mg once daily)	Ritonavir: Morning AUC: ↑ 18% (↑ 6% to ↑ 33%) Evening AUC: ↔ Morning C_{min} : ↑ 24% (↑ 12% to ↑ 38%) Evening C_{min} : ↔ Morning C_{min} : ↑ 42% (↑ 9% to ↑ 86%) ^b Evening C_{min} : ↑ 24% (↑ 3% to ↑ 50%) ^b Efavirenz: AUC: ↑ 21% (↑ 10% to ↑ 34%) C_{min} : ↑ 14% (↑ 4% to ↑ 26%) C_{min} : ↑ 25% (↑ 7% to ↑ 46%) ^b (inhibition of CYP-mediated oxidative metabolism) When efavirenz was given with ritonavir 500 mg or 600 mg twice daily, the combination was not well tolerated (for example, dizziness, nausea, paraesthesia and elevated liver enzymes occurred). Sufficient data on the tolerability of efavirenz with low-dose ritonavir (100 mg, once or twice daily) are not available.	When using efavirenz with low-dose ritonavir, the possibility of an increase in the incidence of efavirenz-associated adverse events should be considered, due to possible pharmacodynamic interaction.
Saquinavir / ritonavir / Efavirenz	Interaction not studied.	No data are available to make a dose recommendation. See also ritonavir row above. Use of efavirenz in combination with saquinavir as the sole protease inhibitor is not recommended.

NRTIs and NNRTIs		
NRTIs / Efavirenz	Specific interaction studies have not been performed with efavirenz and NRTIs other than lamivudine, zidovudine, and tenofovir disoproxil fumarate. Clinically significant interactions are not expected since the NRTIs are metabolised via a different route than efavirenz and would be unlikely to compete for the same metabolic enzymes and elimination pathways.	No dosage adjustment is necessary for either medicinal product.
NNRTIs/Efavirenz	Interaction not studied.	Since use of two NNRTIs proved not beneficial in terms of efficacy and safety, co-administration of efavirenz and another NNRTI is not recommended.
Antibiotics		
Azithromycin / Efavirenz (600 mg single dose / 400 mg once daily)	No clinically significant pharmacokinetic interaction.	No dosage adjustment is necessary for either medicinal product.
Clarithromycin / Efavirenz (500 mg q12h / 400 mg once daily)	Clarithromycin: AUC: ↓ 39% (↓ 30% to ↓ 46%) C_{min} : ↓ 26% (↓ 15% to ↓ 35%) Clarithromycin 14-hydroxymetabolite: AUC: ↑ 34% (↑ 18% to ↑ 53%) C_{min} : ↑ 49% (↑ 32% to ↑ 69%) Efavirenz: AUC: ↔ C_{min} : ↑ 11% (↑ 3% to ↑ 19%) (CYP3A4 induction) Rash developed in 46% of uninfected volunteers receiving efavirenz and clarithromycin.	The clinical significance of these changes in clarithromycin plasma levels is not known. Alternatives to clarithromycin (e.g. azithromycin) may be considered. No dosage adjustment is necessary for efavirenz.
Other macrolide antibiotics (e.g., erythromycin) / Efavirenz	Interaction not studied.	No data are available to make a dose recommendation.
Antimycobacterials		
Rifabutin / Efavirenz (300 mg once daily / 600 mg once daily)	Rifabutin: AUC: ↓ 38% (↓ 28% to ↓ 47%) C_{min} : ↓ 32% (↓ 15% to ↓ 46%) C_{min} : ↓ 45% (↓ 31% to ↓ 56%) Efavirenz: AUC: ↔ C_{min} : ↔ C_{min} : ↓ 12% (↓ 24% to ↑ 1%) (CYP3A4 induction)	The daily dose of rifabutin should be increased by 50% when administered with efavirenz. Consider doubling the rifabutin dose in regimens where rifabutin is given 2 or 3 times a week in combination with efavirenz.
Rifampicin / Efavirenz (600 mg once daily / 600 mg once daily)	Efavirenz: AUC: ↓ 26% (↓ 15% to ↓ 36%) C_{min} : ↓ 20% (↓ 11% to ↓ 28%) C_{min} : ↓ 32% (↓ 15% to ↓ 46%) (CYP3A4 and CYP2B6 induction)	When taken with rifampicin, increasing efavirenz daily dose to 800 mg may provide exposure similar to a daily dose of 600 mg when taken without rifampicin. The clinical effect of this dose adjustment has not been adequately evaluated. Individual tolerability and virological response should be considered when making the dose adjustment. No dosage adjustment is necessary for rifampicin.
Antifungals		
Itraconazole / Efavirenz (200 mg q12h / 600 mg once daily)	Itraconazole: AUC: ↓ 39% (↓ 21% to ↓ 53%) C_{min} : ↓ 37% (↓ 20% to ↓ 51%) C_{min} : ↓ 44% (↓ 27% to ↓ 58%) (decrease in itraconazole concentrations: CYP3A4 induction) Hydroxyitraconazole: AUC: ↓ 37% (↓ 14% to ↓ 55%) C_{min} : ↓ 35% (↓ 12% to ↓ 52%) C_{min} : ↓ 43% (↓ 18% to ↓ 60%) Efavirenz: No clinically significant pharmacokinetic change.	Since no dose recommendation for itraconazole can be made, alternative antifungal treatment should be considered.
Voriconazole / Efavirenz (200 mg twice daily / 400 mg once daily)	Voriconazole: AUC: ↓ 77% C_{min} : ↓ 61% Efavirenz: AUC: ↓ 44% C_{min} : ↓ 38% Voriconazole: AUC: ↑ 7% (↓ 23% to ↑ 13%)* C_{min} : ↑ 23% (↓ 1% to ↑ 53%)* Efavirenz: AUC: ↑ 17% (↑ 6% to ↑ 29%)* C_{min} : ↔** **compared to 200 mg twice daily alone **compared to 600 mg once daily alone (competitive inhibition of oxidative metabolism)	When efavirenz is co-administered with voriconazole, the voriconazole maintenance dose must be increased to 400 mg twice daily and the efavirenz dose must be reduced by 50%, i.e., to 300 mg once daily. When treatment with voriconazole is stopped, the initial dosage of efavirenz should be restored.
Voriconazole / Efavirenz (400 mg twice daily / 300 mg once daily)	Voriconazole: AUC: ↑ 7% (↓ 23% to ↑ 13%)* C_{min} : ↑ 23% (↓ 1% to ↑ 53%)* Efavirenz: AUC: ↑ 17% (↑ 6% to ↑ 29%)* C_{min} : ↔** **compared to 200 mg twice daily alone **compared to 600 mg once daily alone (competitive inhibition of oxidative metabolism)	
Fluconazole / Efavirenz (200 mg once daily / 400 mg once daily)	No clinically significant pharmacokinetic interaction	No dosage adjustment is necessary for either medicinal product.
Ketoconazole and other imidazole antifungals	Interaction not studied	No data are available to make a dose recommendation.
ACID REDUCING AGENTS		
Aluminium hydroxide- magnesium hydroxide-simethicone antacid / Efavirenz (30 mL single dose / 400 mg single dose) Famotidine / Efavirenz (40 mg single dose / 400 mg single dose)	Neither aluminium / magnesium hydroxide antacid nor famotidine altered the absorption of efavirenz.	Co-administration of efavirenz with medicinal products that alter gastric pH would not be expected to affect efavirenz absorption.
ANTI-ANXIETY AGENTS		
Lorazepam / Efavirenz (2 mg single dose / 600 mg once daily)	Lorazepam: AUC: ↑ 7% (↑ 1% to ↑ 14%) C_{min} : ↑ 16% (↑ 2% to ↑ 32%) These changes are not considered clinically significant.	No dosage adjustment is necessary for either medicinal product.
ANTICONSULSANTS		
Carbamazepine / Efavirenz (400 mg once daily / 600 mg once daily)	Carbamazepine: AUC: ↓ 27% (↓ 20% to ↓ 33%) C_{min} : ↓ 20% (↓ 15% to ↓ 24%) C_{min} : ↓ 35% (↓ 24% to ↓ 44%) Efavirenz: AUC: ↓ 36% (↓ 32% to ↓ 40%) C_{min} : ↓ 21% (↓ 15% to ↓ 26%) C_{min} : ↓ 47% (↓ 41% to ↓ 53%) (decrease in carbamazepine concentrations: CYP3A 4 induction; decrease in efavirenz concentrations: CYP3A4 and CYP2B6 induction) The steady-state AUC, C_{min} and C_{max} of the active carbamazepine epoxide metabolite remained unchanged. Co-administration of higher doses of either efavirenz or carbamazepine has not been studied.	No dosage recommendation can be made. An alternative anticonvulsant should be considered. Carbamazepine plasma levels should be monitored periodically.

 Artwork Implementation Schedule Check (✓) whichever is applicable (Approval is not valid without following details)		Date of Issue		Issued By										
Packaging Development		Date of Return												
		Material Code	75091073	Supersedes	75057083	Market	Malaysia							
		Description	LIT. EFAMAT TABS 600 mg (36 Months) MY V2											
		Component	Printed Literature	Actual Size	Flat: 360 x 380 mm; Folded: 34 x 50 mm)									
		Substrate	60 gsm Century Maplitho Paper											
		Design & Style	Supply in Folded form as Proposed Size (with tape)											
		Reason for Issue	Change in Text & Size											
		Printing Pantone Nos	1	BLACK	2	NA	3	NA	4	NA				
			5	NA	6	NA	7	NA	8	NA				
		Non Printing		Die Line	O	NA	O	NA	O	NA				
Proof No.	1	2	3	4	5	Prepared By		Checked By		Approved By				
Remarks	Revised a/w	corrections	x	x	x	Packaging Development		Packaging Development		Production				
	05.07; 22.08; 30.08; 06.07.09;													
SOP-000565164-FORM-000565208-A01-03-01-20			Final Date		dd/mm/yy									

Phenytoin, Phenobarbital, and other anticonvulsants that are substrates of CYP450 isoenzymes	Interaction not studied. There is a potential for reduction or increase in the plasma concentrations of phenytoin, phenobarbital and other anticonvulsants that are substrates of CYP450 isoenzymes when co-administered with efavirenz.	When efavirenz is co-administered with an anticonvulsant that is a substrate of CYP450 isoenzymes, periodic monitoring of anticonvulsant levels should be conducted.
Vigabatrin / Efavirenz Gabapentin / Efavirenz	Interaction not studied. Clinically significant interactions are not expected since vigabatrin and gabapentin are exclusively eliminated unchanged in the urine and are unlikely to compete for the same metabolic enzymes and elimination pathways as efavirenz.	No dosage adjustment is necessary for any of these medicinal products.
ANTI DEPRESSANTS		
Selective Serotonin Reuptake Inhibitors (SSRIs)		
Sertraline / Efavirenz (50 mg once daily / 600 mg once daily)	Sertraline: AUC: ↓ 39% (↓ 27% to ↓ 50%) C _{max} : ↓ 29% (↓ 15% to ↓ 40%) C _{min} : ↓ 46% (↓ 31% to ↓ 58%) Efavirenz: AUC: ↔ C _{max} : ↑ 11% (↑ 6% to ↑ 16%) C _{min} : ↔ (CYP3A4 induction)	Sertraline dose increases should be guided by clinical response. No dosage adjustment is necessary for efavirenz.
Paroxetine / Efavirenz (20 mg once daily / 600 mg once daily)	No clinically significant pharmacokinetic interaction.	No dosage adjustment is necessary for either medicinal product.
Fluoxetine / Efavirenz	Interaction not studied. Since fluoxetine shares a similar metabolic profile with paroxetine, i.e. a strong CYP2D6 inhibitory effect, a similar lack of interaction would be expected for fluoxetine.	No dosage adjustment is necessary for either medicinal product.
ANTI HISTAMINES		
Cetirizine / Efavirenz (10 mg single dose / 600 mg once daily)	Cetirizine: AUC: ↔ C _{max} : ↓ 24% (↓ 18% to ↓ 30%) These changes are not considered clinically significant. Efavirenz: No clinically significant pharmacokinetic interaction.	No dosage adjustment is necessary for either medicinal product.
CARDIOVASCULAR AGENTS		
Calcium Channel Blockers		
Diltiazem / Efavirenz (240 mg once daily / 600 mg once daily)	Diltiazem: AUC: ↓ 69% (↓ 55% to ↓ 79%) C _{max} : ↓ 60% (↓ 50% to ↓ 68%) C _{min} : ↓ 63% (↓ 44% to ↓ 75%) Desacetyl diltiazem: AUC: ↓ 75% (↓ 59% to ↓ 84%) C _{max} : ↓ 64% (↓ 57% to ↓ 69%) C _{min} : ↓ 62% (↓ 44% to ↓ 75%) N-monodemethyl diltiazem: AUC: ↓ 37% (↓ 17% to ↓ 52%) C _{max} : ↓ 28% (↓ 7% to ↓ 44%) C _{min} : ↓ 37% (↓ 17% to ↓ 52%) Efavirenz: AUC: ↑ 11% (↑ 5% to ↑ 18%) C _{max} : ↑ 16% (↑ 6% to ↑ 26%) C _{min} : ↑ 13% (↑ 1% to ↑ 26%) (CYP3A4 induction) The increase in efavirenz pharmacokinetic parameters is not considered clinically significant.	Dose adjustments of diltiazem should be guided by clinical response (refer to the Summary of Product Characteristics for diltiazem). No dosage adjustment is necessary for efavirenz.
Verapamil, Felodipine, Nifedipine and Nidardipine	Interaction not studied. When efavirenz is co-administered with a calcium channel blocker that is a substrate of the CYP3A4 enzyme, there is a potential for reduction in the plasma concentrations of the calcium channel blocker.	Dose adjustments of calcium channel blockers should be guided by clinical response (refer to the Summary of Product Characteristics for the calcium channel blocker).
LIPID LOWERING MEDICINAL PRODUCTS		
HMG Co-A Reductase Inhibitors		
Atorvastatin / Efavirenz (10 mg once daily / 600 mg once daily)	AUC: ↓ 43% (↓ 34% to ↓ 50%) C _{max} : ↓ 12% (↓ 1% to ↓ 26%) 2-hydroxy atorvastatin: AUC: ↓ 35% (↓ 13% to ↓ 40%) C _{max} : ↓ 13% (↓ 0% to ↓ 23%) 4-hydroxy atorvastatin: AUC: ↓ 4% (↓ 0% to ↓ 31%) C _{max} : ↓ 47% (↓ 9% to ↓ 51%) Total active HMG Co-A reductase inhibitors: AUC: ↓ 34% (↓ 21% to ↓ 41%) C _{max} : ↓ 20% (↓ 2% to ↓ 26%)	Cholesterol levels should be periodically monitored. Dosage adjustment of atorvastatin may be required (refer to the Summary of Product Characteristics for atorvastatin). No dosage adjustment is necessary for efavirenz.
Pravastatin / Efavirenz (40 mg once daily / 600 mg once daily)	Pravastatin: AUC: ↓ 40% (↓ 26% to ↓ 57%) C _{max} : ↓ 18% (↓ 59% to ↑ 12%)	Cholesterol levels should be periodically monitored. Dosage adjustment of pravastatin may be required (refer to the Summary of Product Characteristics for pravastatin). No dosage adjustment is necessary for efavirenz.
Simvastatin / Efavirenz (40 mg once daily / 600 mg once daily)	AUC: ↓ 69% (↓ 62% to ↓ 73%) C _{max} : ↓ 76% (↓ 63% to ↓ 79%) Simvastatin acid: AUC: ↓ 58% (↓ 39% to ↓ 68%) C _{max} : ↓ 51% (↓ 32% to ↓ 58%) Total active HMG Co-A reductase inhibitors: AUC: ↓ 60% (↓ 52% to ↓ 68%) C _{max} : ↓ 62% (↓ 55% to ↓ 78%) Total HMG Co-A reductase inhibitors: AUC: ↓ 60% (↓ 54% to ↓ 74%) C _{max} : ↓ 70% (↓ 54% to ↓ 85%) (CYP3A4 induction) Co-administration of efavirenz with atorvastatin, pravastatin, or simvastatin did not affect efavirenz AUC or C _{max} values.	Cholesterol levels should be periodically monitored. Dosage adjustment of simvastatin may be required (refer to the Summary of Product Characteristics for simvastatin). No dosage adjustment is necessary for efavirenz.
HORMONAL CONTRACEPTIVES		
Ethinylloestradiol / Efavirenz (50 µg single dose / 400 mg once daily)	Ethinylloestradiol: AUC: ↑ 37% (↑ 25% to ↑ 51%) C _{max} : ↔ Efavirenz: AUC: ↔ C _{max} : ↔ (mechanism unknown) The clinical significance of the effect on ethinylloestradiol AUC is not known. Only the ethinylloestradiol component of oral contraceptives has been studied.	The potential interaction of efavirenz with oral contraceptives has not been fully characterised. A reliable method of barrier contraception must be used in addition to oral contraceptives.

IMMUNOSUPPRESSANTS		
Immunosuppressants metabolized by CYP3A4 (eg, cyclosporine, tacrolimus, sirolimus) / Efavirenz	Interaction not studied. Decreased exposure of the immunosuppressant may be expected (CYP3A4 induction). These immunosuppressants are not anticipated to affect exposure of efavirenz.	Dose adjustments of the immunosuppressant may be required. Close monitoring of immunosuppressant concentrations for at least 2 weeks (until stable concentrations are reached) is recommended when starting or stopping treatment with efavirenz.
OPIOIDS		
Methadone / Efavirenz (stable maintenance, 35-100 mg once daily / 600 mg once daily)	Methadone: AUC: ↓ 52% (↓ 33% to ↓ 66%) C _{max} : ↓ 45% (↓ 25% to ↓ 59%) (CYP3A4 induction) In a study of HIV infected intravenous drug users, co-administration of efavirenz with methadone resulted in decreased plasma levels of methadone and signs of opiate withdrawal. The methadone dose was increased by a mean of 22% to alleviate withdrawal symptoms.	Patients should be monitored for signs of withdrawal and their methadone dose increased as required to alleviate withdrawal symptoms.

^a 90% confidence intervals unless otherwise noted.
^b 95% confidence intervals.

Pregnancy and lactation
Efavirenz should not be used during pregnancy unless there are no other appropriate treatment options.
Women of childbearing potential: Pregnancy should be avoided in women treated with efavirenz. Barrier contraception should always be used in combination with other methods of contraception (for example, oral or other hormonal contraceptives). Because of the long half-life of efavirenz, use of adequate contraceptive measures for 12 weeks after discontinuation of efavirenz is recommended. Women of child bearing potential should undergo pregnancy testing before initiation of efavirenz.
Pregnancy: There are no adequate and well-controlled studies of efavirenz in pregnant women. In postmarketing experience through an antiretroviral pregnancy registry, more than 200 pregnancies with first-trimester exposure to efavirenz as part of a combination antiretroviral regimen have been reported with no specific malformation pattern. Retrospectively in this registry, a small number of cases of neural tube defects, including meningomyelocele, have been reported but causality has not been established. Studies in animals have shown reproductive toxicity including marked teratogenic effects.
Lactation: Studies in rats have demonstrated that efavirenz is excreted in milk reaching concentrations much higher than those in maternal plasma. It is not known whether efavirenz is excreted in human milk. Since animal data suggest that the substance may be passed into breast milk, it is recommended that mothers taking efavirenz do not breast feed their infants. It is recommended that HIV infected women do not breast feed their infants under any circumstances in order to avoid transmission of HIV.

Effects on ability to drive and use machines
Efavirenz has not been specifically evaluated for possible effects on the ability to drive a car or operate machinery. Efavirenz may cause dizziness, impaired concentration, and/or somnolence. Patients should be instructed that if they experience these symptoms they should avoid potentially hazardous tasks such as driving or operating machinery.
Undesirable effects
The most frequently reported treatment-related undesirable effects of at least moderate severity reported in patients were rash, dizziness, nausea, headache and fatigue. The most notable undesirable effects associated with efavirenz are rash and nervous system symptoms. The administration of Efavirenz with food may increase efavirenz exposure and may lead to an increase in the frequency of undesirable effects. Long-term use of efavirenz was not associated with any new safety concerns.
Rash: Rashes are usually mild to moderate maculopapular skin eruptions that occur within the first two weeks of initiating therapy with efavirenz. In most patients rash resolves with continuing therapy with efavirenz within one month. Efavirenz can be reinitiated in patients interrupting therapy because of rash. Use of appropriate antihistamines and/or corticosteroids is recommended when efavirenz is restarted. Experience with efavirenz in patients who discontinued other antiretroviral agents of the NNRTI class is limited.
Psychiatric symptoms: Serious psychiatric adverse experiences have been reported in patients treated with efavirenz. The specific serious psychiatric events are detailed hereafter:
— severe depression
— suicidal ideation
— non-fatal suicide attempts
— aggressive behaviour
— paranoid reactions
— manic reactions

Patients with a history of psychiatric disorders appear to be at greater risk of these serious psychiatric adverse experiences with the frequency of each of the above events ranging from 0.3% for manic reactions to 2.0% for both severe depression and suicidal ideation. There have also been post marketing reports of death by suicide, delusions and psychosis-like behaviour.

Nervous system symptoms: The frequently reported undesirable effects in patients receiving 600 mg efavirenz with other antiretroviral agents included, but were not limited to: dizziness, insomnia, somnolence, impaired concentration and abnormal dreaming. Nervous system symptoms of moderate-to-severe intensity were experienced by 19.4% of patients compared to 9.0% of patients receiving control regimens. These symptoms were severe in 2.0% of patients receiving efavirenz 600 mg daily and in 1.3% of patients receiving control regimens. In clinical studies 2.1% of patients treated with 600 mg of efavirenz discontinued therapy because of nervous system symptoms.

Nervous system symptoms usually began during the first one or two days of therapy and generally resolve after the first 2-4 weeks.
Nervous system symptoms may occur more frequently when efavirenz is taken concomitantly with meals possibly due to increased efavirenz plasma levels. Dosing at bedtime seems to improve the tolerability of these symptoms and can be recommended during the first weeks of therapy and in patients who continue to experience these symptoms. Dose reduction or splitting the daily dose has not been shown to provide benefit.

Adverse reactions of moderate or greater severity with at least possible relationship to treatment regimen (based on investigator attribution) reported in clinical trials of efavirenz at the recommended dose in combination therapy (n = 1,008) are listed below. Frequency is defined using the following convention: very common (≥ 1/100); common (≥ 1/100, < 1/10); uncommon (≥ 1/1,000, < 1/100); rare (≥ 1/10,000, < 1/1,000); very rare (< 1/10,000) including isolated reports.

Immune system disorders
Uncommon: hypersensitivity
Psychiatric disorders
Common: anxiety, depression
Uncommon: affect lability, aggression, euphoric mood, hallucination, mania, paranoia, suicide attempt, suicide ideation
Nervous system disorders
Common: abnormal dreams, disturbance in attention, dizziness, headache, insomnia, somnolence
Uncommon: agitation, amnesia, ataxia, coordination abnormal, confusional state, convulsions, thinking abnormal
Eye disorders
Uncommon: vision blurred
Ear and labyrinth disorders
Uncommon: vertigo
Gastrointestinal disorders
Common: abdominal pain, diarrhoea, nausea, vomiting
Uncommon: pancreatitis acute
Hepatobiliary disorders
Uncommon: hepatitis acute
Skin and subcutaneous tissue disorders
Very common: rash
Common: pruritus
Uncommon: erythema multiforme, Stevens-Johnson syndrome
General disorders and administration site conditions
Common: fatigue
Reproductive system and breast disorders
Uncommon: gynaecomastia
Immune Reactivation Syndrome: In HIV infected patients with severe immune deficiency at the time of initiation of combination antiretroviral therapy (CART), an inflammatory reaction to opportunistic or residual opportunistic infections may arise.
Lipodystrophy and metabolic abnormalities: Combination antiretroviral therapy has been associated with redistribution of body fat (lipodystrophy) in HIV patients including the loss of peripheral and facial subcutaneous fat, increased intra-abdominal and visceral fat, breast hypertrophy and dorsocervical fat accumulation (buffalo hump).
Combination antiretroviral therapy has been associated with metabolic abnormalities such as hypertriglyceridaemia, hypercholesterolaemia, insulin resistance, hyperglycaemia and hyperlactataemia.
Osteonecrosis: Cases of osteonecrosis have been reported, particularly in patients with generally acknowledged risk factors, advanced HIV disease or long-term exposure to combination antiretroviral therapy (CART). The frequency of this is unknown.
Cannabinoid test interaction: Efavirenz does not bind to cannabinoid receptors. False positive urine cannabinoid test results have been reported in uninfected volunteers who received efavirenz. False positive test results have only been observed with the CEDA DAU Multi-Level THC assay, which is used for screening, and have not been observed with other cannabinoid assays tested including tests used for confirmation of positive results.

Post marketing experience
Encephalopathy

Post marketing experience with efavirenz has shown the following additional adverse events to occur in association with efavirenz-containing antiretroviral treatment regimens: delusion, gynaecomastia, hepatic failure, neurosis, photoallergic dermatitis, psychosis and completed suicide.

Adolescents and children: Undesirable effects in children were generally similar to those of adult patients.

Rash was reported more frequently in children and was more often of higher grade than in adults.

Prophylaxis with appropriate antihistamines prior to initiating therapy with efavirenz in children may be considered. Although nervous system symptoms are difficult for young children to report, they appear to be less frequent in children and were generally mild.

Overdose

Some patients accidentally taking 600 mg twice daily have reported increased nervous system symptoms. One patient experienced involuntary muscle contractions.

Treatment of overdose with efavirenz should consist of general supportive measures, including monitoring of vital signs and observation of the patient's clinical status. Administration of activated charcoal may be used to aid removal of unabsorbed efavirenz. There is no specific antidote for overdose with efavirenz. Since efavirenz is highly protein bound, dialysis is unlikely to remove significant quantities of it from blood.

Pharmacodynamic properties

Pharmacotherapeutic group: HIV-1 specific NNRTI.

ATC code: J05A G03

Mechanism of action: Efavirenz is a NNRTI of HIV-1. Efavirenz is a non-competitive inhibitor of HIV-1 reverse transcriptase (RT) and does not significantly inhibit HIV-2 RT or cellular DNA polymerases (α, β, γ or δ).

Antiviral activity: The free concentration of efavirenz reported for 90 to 95% inhibition of wild type or zidovudine resistant laboratory and clinical isolates *in vitro* ranged from 0.46 to 6.8 nM in lymphoblastoid cell lines, peripheral blood mononuclear cells (PBMCs) and macrophage/monocyte cultures.

Resistance: The potency of efavirenz in cell culture against viral variants with amino acid substitutions at positions 48, 108, 179, 181 or 236 in RT or variants with amino acid substitutions in the protease was similar to that observed against wild type viral strains. The single substitutions which led to the highest resistance to efavirenz in cell culture correspond to a leucine to isoleucine change at position 100 (L100I, 17 to 22-fold resistance) and a lysine-to-asparagine at position 103 (K103N, 18 to 33 fold resistance). Greater than 100 fold loss of susceptibility was observed against HIV variants expressing K103N in addition to other amino acid substitutions in RT.

K103N was the most frequently observed RT substitution in viral isolates from patients who experienced a significant rebound in viral load during clinical studies of efavirenz in combination with didanosine or zidovudine + lamivudine. This mutation was observed in 90% of patients receiving efavirenz with virological failure. Substitutions at RT positions 98, 100, 101, 108, 138, 188, 190 or 225 were also observed, but at lower frequencies, and often only in combination with K103N. The pattern of amino acid substitutions in RT associated with resistance to efavirenz was independent of the other antiviral medications used in combination with efavirenz.

Cross resistance: Cross resistance profiles for efavirenz, nevirapine and delavirdine in cell culture demonstrated that the K103N substitution confers loss of susceptibility to all three NNRTIs. Two of three delavirdine-resistant clinical isolates examined were cross-resistant to efavirenz and contained the K103N substitution. A third isolate which carried a substitution at position 236 of RT was not cross-resistant to efavirenz. Viral isolates recovered from PBMCs of patients enrolled in efavirenz clinical studies who showed evidence of treatment failure (viral load rebound) were assessed for susceptibility to NNRTIs. Thirteen isolates previously characterised as efavirenz-resistant were also resistant to nevirapine and delavirdine. Five of these NNRTI-resistant isolates were found to have K103N or a valine-to-isoleucine substitution at position 108 (V108I) in RT. Three of the efavirenz treatment failure isolates tested remained sensitive to efavirenz in cell culture and were also sensitive to nevirapine and delavirdine.

The potential for cross resistance between efavirenz and PIs is low because of the different enzyme targets involved. The potential for cross-resistance between efavirenz and NRTIs is low because of the different binding sites on the target and mechanism of action.

Pharmacodynamic effects

Efavirenz has not been studied in controlled studies in patients with advanced HIV disease, namely with CD4 counts < 50 cells/mm³, or in PI or NNRTI experienced patients. Clinical experience in controlled studies with combinations including didanosine or zalcitabine is limited.

Pharmacokinetic properties

Absorption

Peak efavirenz plasma concentrations of 1.6 to 9.1 µM were attained by 5 hours following single oral doses of 100 mg to 1,600 mg administered to uninfected volunteers. Dose-related increases in C_{max} and AUC were seen for doses up to 1600 mg; the increases were less than proportional suggesting diminished absorption at higher doses. Time to peak plasma concentrations (3-5 hours) did not change following multiple dosing and steady-state plasma concentrations were reached in 6-7 days.

In HIV infected patients at steady state, mean C_{max}, mean C_{min} and mean AUC were linear with 200 mg, 400 mg, and 600 mg daily doses. In 35 patients receiving efavirenz 600 mg once daily, steady state C_{max} was 12.9 ± 3.7 µM (29%) [Mean ± S.D. (% C.V.)], steady state C_{min} was 5.6 ± 3.2 µM (57%), and AUC was 184 ± 73 µM•h (40%).

Effect of food: The AUC and C_{max} of a single 600 mg dose of efavirenz film-coated tablets in uninfected volunteers was increased by 28% (90% CI: 22 - 33%) and 79% (90% CI: 58 - 102%), respectively, when given with a high fat meal, relative to when given under fasted conditions.

Distribution: Efavirenz is highly bound (approximately 99.5 - 99.75%) to human plasma proteins, predominantly albumin. In HIV-1 infected patients (n = 9) who received efavirenz 200 to 600 mg once daily for at least one month, cerebrospinal fluid concentrations ranged from 0.26 to 1.19% (mean 0.69%) of the corresponding plasma concentration. This proportion is approximately 3-fold higher than the non-protein-bound (free) fraction of efavirenz in plasma.

Biotransformation: Studies in humans and *in vitro* studies using human liver microsomes have demonstrated that efavirenz is principally metabolised by the cytochrome P450 system to hydroxylated metabolites with subsequent glucuronidation of these hydroxylated metabolites. These metabolites are essentially inactive against HIV-1. The *in vitro* studies suggest that CYP3A4 and CYP2B6 are the major isozymes responsible for efavirenz metabolism and that it inhibited P450 isozymes 2C9, 2C19, and 3A4. In *in vitro* studies efavirenz did not inhibit CYP2E1 and inhibited CYP2D6 and CYP1A2 only at concentrations well above those achieved clinically.

Efavirenz plasma exposure may be increased in patients with the homozygous G516T genetic variant of the CYP2B6 isoenzyme. The clinical implications of such an association are unknown; however, the potential for an increased frequency and severity of efavirenz-associated adverse events cannot be excluded.

Efavirenz has been shown to induce P450 enzymes, resulting in the induction of its own metabolism. In uninfected volunteers, multiple doses of 200 - 400 mg per day for 10 days resulted in a lower than predicted extent of accumulation (22 - 42% lower) and a shorter terminal half-life compared with single dose administration (see below).

Elimination: Efavirenz has a relatively long terminal half-life of 52 to 76 hours after single doses and 40 - 55 hours after multiple doses. Approximately 14 - 34% of a radiolabelled dose of efavirenz was recovered in the urine and less than 1% of the dose was excreted in urine as unchanged efavirenz.

In the single patient studied with severe hepatic impairment (Child Pugh Grade C), half life was doubled indicating a potential for a much greater degree of accumulation.

Cardiac Electrophysiology

The effect of Efavirenz Tablets on the QTc interval was evaluated in an open-label, positive and placebo controlled, fixed single sequence 3-period, 3-treatment crossover QT study in 58 healthy subjects enriched for CYP2B6 polymorphisms. The mean C_{max} of efavirenz in subjects with CYP2B6 *6/*6 genotype following the administration of 600mg daily dose for 14 days was 2.25-fold the mean C_{max} observed in subjects with CYP2B6 *1/*1 genotype. A positive relationship between efavirenz concentration and QTc prolongation was observed. Based on the concentration-QTc relationship, the mean QTc prolongation and its upper bound 90% confidence interval are 8.7 ms and 11.3 ms in subjects with CYP2B6*6/*6 genotype following the administration of 600mg daily dose for 14 days. (see Section Warnings and Precautions & Section Interaction with Other Medicaments).

Paediatric pharmacokinetics: In 49 paediatric patients receiving the equivalent of a 600 mg dose of efavirenz (dose adjusted from calculated body size based on weight), steady state C_{max} was 14.1 µM, steady state C_{min} was 5.6 µM, and AUC was 216 µM•h. The pharmacokinetics of efavirenz in paediatric patients were similar to adults.

Gender, race, elderly: Pharmacokinetics of efavirenz in patients appear to be similar between men and women and among the racial groups studied. Although limited data suggest that Asian and Pacific Island patients may have higher exposure to efavirenz, they do not appear to be less tolerant of efavirenz. Pharmacokinetic studies have not been performed in the elderly.

Shell life 36 Months	
Special precautions for storage Do not store above 30°C. Store in the original container.	
Pack HDPE bottle of 30's.	

Mfd. by:
Mylan Laboratories Limited
F-4 & F-12, MIDC, Malegaon, Sinnar,
Mylan Nashik - 422 113, Maharashtra, INDIA

Product Registration Holder:
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Jalan Meru, 41050 Klang Selangor. MALAYSIA.

November 2021

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 Mylan Laboratories Limited Packaging Development	Artwork Implementation Schedule Check (✓) whichever is applicable (Approval is not valid without following details)					Date of Issue		Issued By				
	☐ New Component ☐ Immediately (Stock of superseded component to be destroyed, if applicable.) ☐ After consumption of existing (superseded) stock. ☐ Other (Specify)					Date of Return						
						Material Code		75091073	Supersedes	75057083	Market	Malaysia
						Description		LIT. EFAMAT TABS 600 mg (36 Months) MY V2				
						Component		Printed Literature		Actual Size	Flat: 360 x 380 mm; Folded: 34 x 50 mm)	
Substrate		60 gsm Century Maplitho Paper										
Design & Style		Supply in Folded form as Proposed Size (with tape)										
Reason for Issue		Change in Text & Size										
Printing		1	BLACK	2	NA	3	NA	4	NA			
Pantone Nos		5	NA	6	NA	7	NA	8	NA			
Non Printing			Die Line		NA		NA		NA			
		Prepared By			Checked By		Approved By					
Packaging Development		Packaging Development		Production		Regulatory Affairs		Quality Assurance				